Z.ai Local Coding Assistant — Read-Only Architecture Discovery Report

This report provides a comprehensive, read-only analysis of the architecture of the **Simple AI Website Builder / Z.ai Local Coding Assistant** repository. It contains codebase evidence, traces flows, catalogs behaviors, lists metrics, maps dependencies, and provides recommendations for the next architecture.

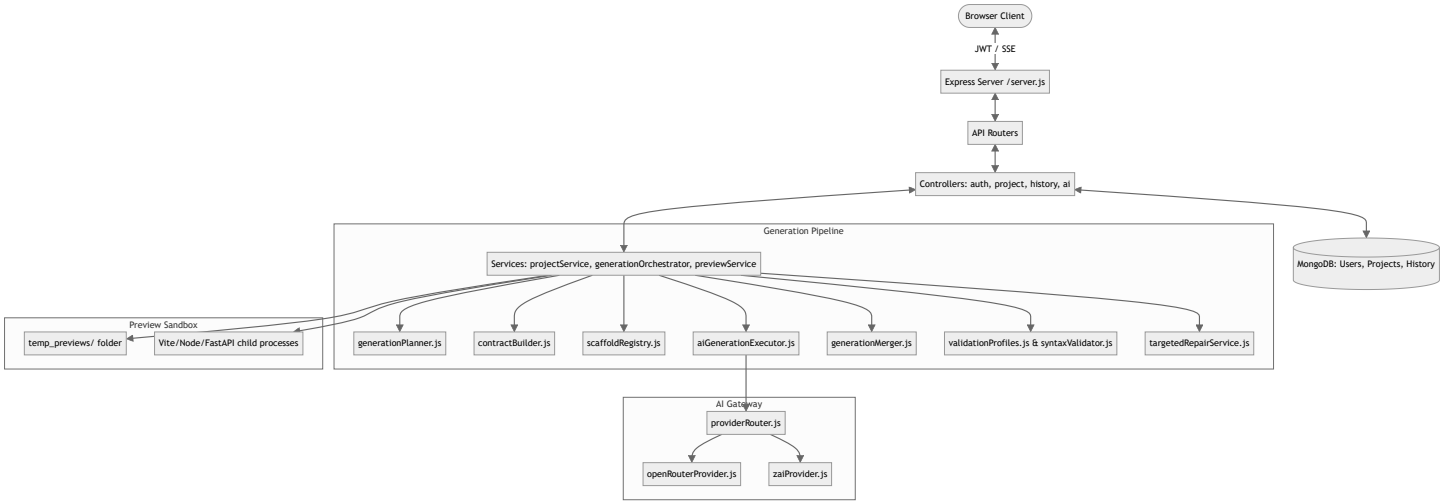
A. Executive Architecture Summary

The **Z.ai Local Coding Assistant** is a developer tool that bridges natural language prompts and locally testable web projects. It acts as an autonomous software agent by:

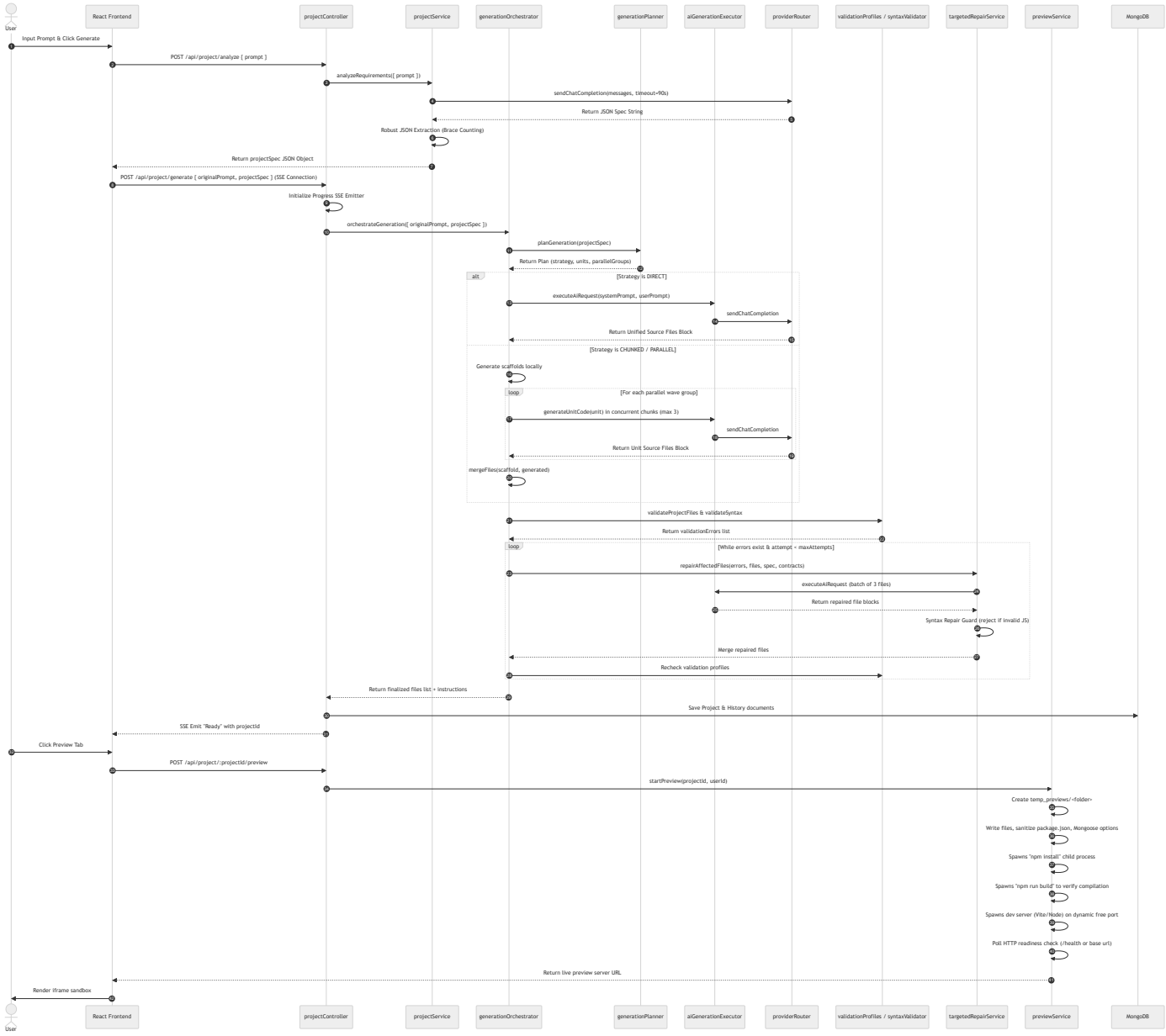
1. Translating a prompt into a structured specification JSON.
2. Generating a local, stack-specific scaffold.
3. Spawning concurrent modular AI calls to write implementation files.
4. Performing multi-phase syntax and completeness validation.
5. Invoking targeted AI repair loops to fix errors.
6. Launching live preview servers within a secure host sandbox.
7. Packaging the project into downloadable ZIP archives.

All data, including code files and specs, are stored in MongoDB. The current backend implementation uses a wave-based scheduling model that supports multiple stacks (MERN, React-Vite, Next.js, Express, FastAPI, Vanilla, and Dynamic Fallback).

B. Current Architecture Diagram



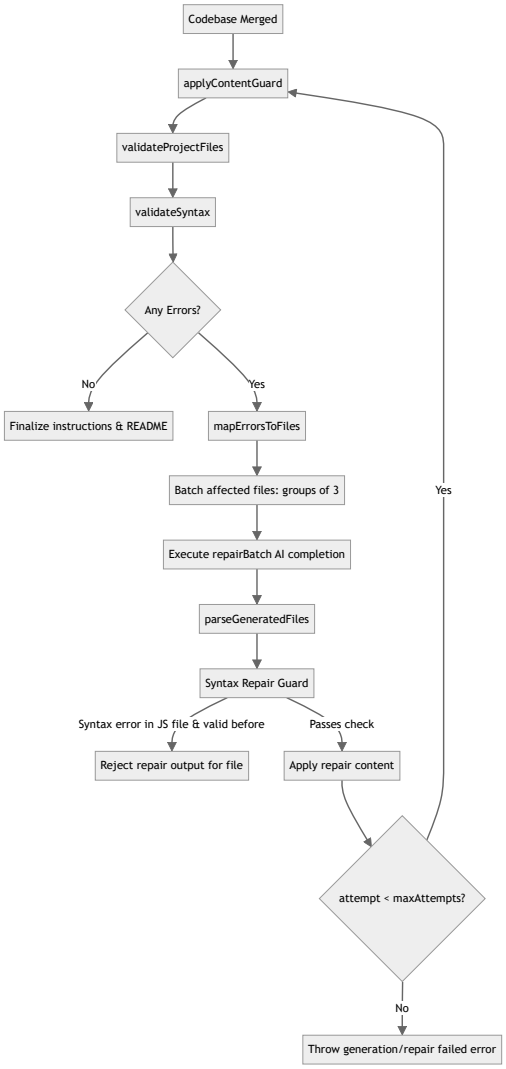
C. Current E2E Execution Sequence Diagram



D. AI Provider Call-Path Diagram

E. Persistence / Recovery Diagram

F. Verification / Repair Flow Diagram



G. Module Dependency Map

The 20 most architecture-critical modules of the Z.ai Local Coding Assistant:

#	Module / Path	Responsibilities	Depends On	Called By
1	server.js	Entry point, DB connection, HTTP Server initialization	db.js, route modules	Node system
2	projectController.js	Maps express requests to projectService, generationOrchestrator, previewService	projectService, generationOrchestrator, previewService, Project, History models	Router
3	projectService.js	Requirements analysis prompt generation, validation wrapper	providerRouter, generationOrchestrator	projectController
4	generationOrchestrator.js	Controls the scaffold/generation/repair pipeline	generationPlanner, scaffoldRegistry, contractBuilder, aiGenerationExecutor, targetedRepairService, validationProfiles	projectController, projectService
5	generationPlanner.js	Strategies, tokens, units, parallelGroups planner	stackProfiles	generationOrches
6	contractBuilder.js	Builds shared contracts folder structures and manifest	stackProfiles	generationOrches
7	scaffoldRegistry.js	local scaffold directory seeding	stackProfiles	generationOrches
8	stackProfiles.js	Contains stack definitions (MERN, Vite, Next, FastAPI)	None	contractBuilder, scaffoldRegistry, generationPlanne previewService

#	Module / Path	Responsibilities	Depends On	Called By
9	aiGenerationExecutor.js	Sends prompts, parses blocks, retries & timeouts calculator	providerRouter	generationOrches targetedRepairSe
10	generationMerger.js	Combines scaffold and generated array lists	None	generationOrches
11	validationProfiles.js	Evaluates file lists, json structures, external deps, imports	contractBuilder, stackProfiles	generationOrches previewService
12	targetedRepairService.js	Maps errors to files and launches repair batches	aiGenerationExecutor, syntaxValidator	generationOrches
13	syntaxValidator.js	Runs Babel parse to verify syntax of JS/JSX	@babel/parser	generationOrches targetedRepairSe previewService
14	previewService.js	Setup temporary workspace, spawn npm install/build/run	Project model, stackProfiles	projectController
15	providerRouter.js	Resolves primary/fallback endpoints and manages failovers	zaiProvider, openRouterProvider	projectService, aiService, aiGenerationExec
16	openRouterProvider.js	OpenRouter axios call client	None	providerRouter
17	zaiProvider.js	Z.ai axios call client	None	providerRouter
18	Project.js	Schema definitions for projects	mongoose	projectController, previewService

#	Module / Path	Responsibilities	Depends On	Called By
19	History.js	Schema definitions for prompt history	mongoose	projectController
20	User.js	Schema definitions for users	mongoose	authController

H. Architecture Gap Matrix

This table lists architectural gaps in the current implementation that prevent scaling up to support arbitrary technologies and larger codebases, mapped to proposed boundary replacements:

Current Architecture Feature / Gap	Architectural Constraint / Problem	Proposed Boundary Replacement
Monolithic requirements prompt	Simple system prompt relies entirely on AI to structure requirements. No conflict detection or stable ID tracking.	`RequirementCompiler` (generates ProjectSpec AST) & `RequirementValidator`
Heuristic execution planner (`generationPlanner.js`)	Fixed conditions and stack checks. High risk of task block failure without ability to retry individual tasks.	`TaskPlanner` & `TaskGraph` (DAG scheduler with topology sort)
Unified in-memory generation array	Entire codebase is held in memory and written at the very end. Process crash resets the whole pipeline.	`FileOperationsEngine` (Virtual VFS) & `CheckpointStore` (granular MongoDB snapshot write)
Dumb, flat context dumping	The entire codebase file list and shared contracts are sent in every AI prompt. bloats token consumption on large codebases.	`ContextBuilder` (assembles only imported files / subgraph files using import parsing)
Hardcoded MERN/stack references	Hardcoded tech profiles in `previewService.js` (ports proxying, env writes, build steps) and `stackProfiles.js` .	`StackAdapter` abstraction framework (seeding commands and adapters out of config files)
Unstructured, monolithic repair	Batches up to 3 files together with error lists. If repair succeeds in file A but fails/breaks file B, it restarts.	`RepairEngine` (targeted single-file repair engine with transactional rollbacks)
Process-direct Live Preview	Launches raw `npm run dev` child processes directly on the host machine using random ports.	Isolated sandboxing or containerized endpoint orchestration

I. Reuse / Wrap / Refactor / Replace Matrix

Recommendations for each architecture-critical subsystem:

Subsystem	Classification	Rationale	Migration Dependency	Migration Risk	Smallest Safe
Provider layer	REFACTOR	`providerRouter.js` and `aiGenerationExecutor.js` work well but have duplicate timeout/retry logic. Refactor into `AIProviderGateway`.	None	Low	Merge the tw blocks into a
Requirement analysis	REFACTOR	The balanced brace parsing is solid. Wrap it inside a `RequirementCompiler` and validate specs using JSON schemas.	None	Low	Add a schem `projectServ`
Planning	REPLACE	Heuristics fail to plan complex graphs. Implement `TaskPlanner` returning a topological `TaskGraph`.	Requirement analysis	Moderate	Let `generati` task map rep custom wave
Generation engine	REFACTOR	Modularize execution units so they are decoupled from the stack profiles structure.	Planning	Moderate	Move MERN- stack profile
Context management	REPLACE	Context size scales quadratically with project files. Context builder must resolve file-dependency imports.	File operations	High	Parse relative generation to code blocks i
File operations	REPLACE	Moving from memory array to transactional VFS enables checkpoints and atomic file mutations.	Persistence	High	Create a `Vir` wrap the gen `generationO`

Subsystem	Classification	Rationale	Migration Dependency	Migration Risk	Smallest Safe
Sandbox	REFACTOR	Process spawning and port management are complex. Decouple them from the file extraction blocks.	None	Moderate	Create a prev in `previewSe`
Verification	REFACTOR	Make verification checks pluggable and modular instead of monolithic inline checks.	None	Low	Create a `Ver` registered ch `DependencyC`
Repair	REPLACE	Batch-repair of 3 files is token-heavy. Move to transactional single-file repair loops.	Verification	Moderate	Modify the re and verify on
Preview	WRAP	Continue using `previewService.js` but wrap it under a unified `PreviewAdapter` boundary.	Sandbox	Low	Define a `Pre`
Persistence	REPLACE	Single MongoDB Project document limits crash recovery. Break down into Tasks, Versions, and Files tables.	File operations	High	Create a Mor `Checkpoints` generation st
Stack support	REPLACE	Decouple stack profiles from hardcoded node/MERN logic. Create plugin-based configuration profiles.	None	Moderate	Convert `stac` configuratior
Artifact packaging	KEEP	`adm-zip` works perfectly.	None	Low	Keep as is.

Subsystem	Classification	Rationale	Migration Dependency	Migration Risk	Smallest Safe
Tests	REFACTOR	Extend unit testing to cover the actual AI providers (using mocks) and requirement parser.	None	Low	Add mocks fo test runner.

J. Large-Project Bottleneck Analysis

Using a large-scale project reference workload (e.g. LearnSphere LMS: courses, lectures, quizzes, profiles, payments, dashboards, instructor portals, analytics):

- **Major Generation Waves / Phases**:

Under the MERN `CHUNKED` strategy, the planner schedules **6 execution waves** (backend_foundation $ → $ backend_api $ → $ frontend_shell $ → $ frontend_pages $ → $ frontend_components $ → $ documentation). These are completely sequential, meaning that any lag in a wave halts the entire pipeline.

- **Likely AI Call Count**:

1 (analysis) + 6 (module generation) + 5 to 15 (repair calls due to the high volume of components and routes). Total AI call count is estimated at **12 to 22 calls** per complete generation.

- **Sequential Bottlenecks**:

All 6 generation phases are sequential. In addition, targeted repairs are processed in sequential batches of 3, meaning that lint errors are fixed slowly, slowing down completion.

- **Context-Growth Bottlenecks**:

As the codebase grows (e.g., 40+ React views and schemas for LearnSphere), the `contracts` object sent in the prompt context grows excessively. The repair loop also passes complete files in the prompt. This will hit the 4000-token prompt budget, causing code truncation or incomplete implementations.

- **Provider Timeout Risks**:

Large generations easily exceed the 90s-240s adaptive timeouts when dealing with high token volumes, causing fake timeout failures and failovers.

- **File-Conflict Risks**:

Since chunks generate components in isolation, there is a high risk that `mergeFiles` overwrites files or imports become broken if they are not generated under absolute contract paths.

- **Verification Cost**:

Running `npm install` and `npm run build` on a complex stack inside the temporary preview sandbox can take 3 to 5 minutes, easily hitting the preview timeout bounds.

- **Repair Amplification Risks**:

Fixing an import error in File A might break File B, which causes a new validation error. In a large project, this leads to repair oscillation, exhausting the 2-attempt limit and failing the entire build.

- **Crash-Recovery Weakness**:

If the server crashes on wave 5, all 4 generated waves are lost from memory. There is no resume checkpoint.

- **Requirement-Loss Risks**:

Without stable requirement IDs, the AI agent in the repair loop might delete complex page logic to fix a React compile error, leading to a successful build that is missing the actual requested features.

K. Minimum-Time Migration Recommendation

Smallest Architecture Change for Reliability

The single most effective and minimal change is **Checkpoint Persistence**.

- **Action**: Instead of keeping files in memory during the orchestrator loop, save the intermediate file list to MongoDB (e.g., in a `Project.draftFiles` field or a temporary state model) after each successful generation unit wave.
- **Why**: If the process crashes or times out, the backend can retrieve the draft state and resume the orchestrator at the next incomplete wave, saving up to 80% of generation time and avoiding double-

billing/wasting tokens.

Top 3 reliability improvements per implementation effort:

1. **Intermediate Checkpoint Saving**: Persists the files list after every generation wave. (Effort: Low, Impact: High)
2. **Syntax & Import Self-Repair Guard**: Validate code syntax using `@babel/parser` *before* files are merged, reverting to the previous state immediately if the AI returns broken syntax. (Effort: Low, Impact: High)
3. **Sub-graph Context Truncation**: Only pass the files/contracts that are directly imported by the current generation unit. (Effort: Moderate, Impact: High)

Postponable Architecture Changes:

- Containerized preview sandboxes (can continue using localized `temp_previews` folders).
- Full AST compilation of user prompts.

L. Ordered Migration Phases

Phase 1: VFS, Checkpoint Store & Gateway Refactor (Weeks 1-2)

- **Dependencies**: None
- **Deliverables**:
- Implement `AIProviderGateway` consolidating provider failovers.
- Introduce `VirtualFileSystem` to manage file writes in memory.
- Add a checkpoint collection in MongoDB. Save files list after each generation wave.

Phase 2: Pluggable Verification & Transactional Repairs (Weeks 3-4)

- **Dependencies**: Phase 1
- **Deliverables**:
- Decouple validation checks into `VerificationEngine`.
- Rewrite `targetedRepairService` to perform single-file repairs with rollbacks if the repair introduces a syntax or import error.

Phase 3: DAG Scheduler & Sub-graph Context Builder (Weeks 5-6)

- **Dependencies**: Phase 2
- **Deliverables**:
- Implement `TaskPlanner` and a true topological `TaskGraph`.
- Introduce `ContextBuilder` that builds prompts containing only imports-relevant contracts/files.

M. Top 10 Risks

1. **Context Exhaustion**: Large projects (LearnSphere) exceeding model token ceilings during repair.
2. **Repair Loops**: Oscillating import errors where fixing one breaks another.
3. **Windows File Locking**: EPERM errors when deleting preview folders, leading to disk bloat.
4. **Preview Timeout**: Package installations taking longer than the allotted 120s–240s preview limit.
5. **Data Inconsistency**: overwriting active code files if two concurrent tasks write to the same path.
6. **Safety Failures**: Models refusing to generate code due to safety filters, triggering unnecessary provider failovers.
7. **Real Secret Leaks**: AI accidentally generating valid API keys in `.env.example`.
8. **Port Collisions**: `getFreePort` race conditions if two previews launch at the exact same millisecond.
9. **Stale Checkpoints**: Resuming a project generation using stale requirement specs.
10. **State Corruption**: If MongoDB crashes mid-write, leaving orphaned preview processes running.

N. Top 10 Unanswered Questions Requiring Runtime Experiments

1. **Vite Port Re-routing**: Can we intercept hot-reload events to route websockets through the dashboard proxy?
2. **GLM Reasoning Limits**: Does Z.ai's reasoning model require system prompt tweaks to prevent reasoning token budget exhaustion?
3. **Windows Process Tree Termination**: Does killing Vite on Windows leave orphaned node/esbuild processes?

4. **Parallel package.json Installations**: Does running concurrent `npm install` inside the same parent environment cause package conflicts?
5. **Repair Batch Size**: Is a batch size of 1 file more reliable and token-efficient than a batch of 3 files?
6. **Babel Parsing Limits**: Does `@babel/parser` support advanced CSS-in-JS or custom file extensions safely?
7. **Dynamic Stack Adapters**: Can we configure a stack (like Ruby on Rails) entirely from JSON schema without compiling JS routes?
8. **Context Window Slicing**: What is the optimal number of lines of file history to send for repair?
9. **Concurrent MongoDB Writes**: Does high client concurrency degrade SSE generation streams?
10. **Sandbox Port Hijacking**: Can preview servers bind to public interfaces instead of `127.0.0.1`?

O. Recommended Phase 1 Scope

We recommend focusing Phase 1 on the following high-priority, low-risk enhancements:

1. **AIProviderGateway Consolidation**: Move openRouter client and zai client under a single interface.
2. **Checkpoint Store Implementation**: Save the generation state after every unit wave. Add a `/resume` endpoint.
3. **VirtualFileSystem Integration**: Abstract file lists inside a VFS class to manage imports validation and paths safety check before writing.

This ensures immediately higher generation success rates with a clear rollback capability.